1. curlybraces mark body of method

2. Every statement must ends with semicolon

3. Java is case sensitive main() & Main()
are different

Comments :-

Comments are used to explain or describe
it

// ⇒ single line command

For longer lines /* */

/** */
↓
Third type of comment

/* */ → not nest in java
if code contains */
code might not deactivate

Java is strongly typed language
Every thing has data type

Hexadecimal 0X

Octal 0

Binary 0b 0x 0B

You can add underscores

1_000_000

human eye only

Java completes removed them

Java does not have unsigned
versions of long, int, short, byte

reduce
complexity

C/C++

long → 4 byte 32bit processor

long → 8byte 64 bit processor

Difficult in making cross platform programs

Java uses fixed size

X: Byte.toUnsignedInt(b)

so
-128 to 127 → 0 to 255

Chance that integer will not be negative

float → 6-7 significant digits → suffix F or f

double → 15 significant digits

$0.125 = 2^{-3}$ can be expressed as $0x1.0p^{-3}$

Floating point -> IEEE 754

Three special floating point values

Overflows & errors

1. NaN
2. positive Infinity
3. negative infinity

for eg:-

$\frac{1}{0} = \infty$ $\frac{0}{0} = NaN$

$-\frac{1}{0} = -\infty$

$$\boxed{\begin{array}{l}\text{Double . POSITIVE\_INFINITY} \\ \text{NEGATIVE\_INFINITY} \\ \text{NAN}\end{array}}$$

Correspondig float

(scare)

You cannot test

- ~~Dou~~ x == Double.NaN

↓

never true

Double.isNaN(x)

✓

Floating point numbers
-> should not be

0.8999
↓
0.9 you would not expect

No precise binary representation of 1/10

precision computations without knowledge -> BigDecimal class

Char Type

Data type stores single character enclosed in single quotes

'A' -> single quote -> char
"A" -> double quote -> string

Originally char -> hold characters

Unicode -> beyond 65536 characters

some character -> 1 char   Latin letters, digits

some -> 2 chars   Rare emoji

Char -> cannot represent every possible

Every Characters

[\u0000 to \uFFFF] → Unicode

\u → escapes work outside

strings & chars

\n → newline

\t → Tab

\\ → Backslash

\" → Double Quote

\' → Single Quote

\b → back-space

CAUTION

Unicode processed first before

Code processes or parses

\u0022 + \u0022

" " + " "

↓

" " → Intended

// \u000A → newline

✗ syntaxError

newline, breaks comment

//

you

in

U

// \u ↓ found

Java ↓ Read 4 hex digits

Syntax Error

You can have any number of \u in Unicode

\uu022

\uuu022

Unicode

\uD800

Add extra u

\uuD800

\u2122 (ASCII)

↓

\u2122

\uD800

So they can convert

ASCII → Unicode

reverse

Unicode → ASCII

reverse use

for eg

\u00e9 → ê

\u00e9

\uD83D\uDE00

↓

fail

sup one